Question Number: 1

Answer:
The OSI model stands for Open Systems Interconnection model. It is a conceptual framework
used to understand network interactions in seven layers.

The seven layers are:
1. Physical - Deals with physical connection
2. Data Link - Handles error detection
3. Network - Manages routing of data
4. Transport - Ensures reliable delivery
5. Session - Manages sessions between applications
6. Presentation - Handles data formatting
7. Application - Provides network services to applications

Question Number: 2

Answer:
TCP/IP is a protocol suite used for internet communication. TCP stands for Transmission
Control Protocol and IP stands for Internet Protocol.

TCP provides reliable, connection-oriented communication. IP handles routing of packets
across networks. Common protocols include HTTP for web browsing and FTP for file transfer.

Question Number: 3

Answer:
Subnetting is the process of dividing a large network into smaller sub-networks. It is
done using subnet masks to create multiple networks from a single IP address range.

For example, if we have network 192.168.1.0/24, we can subnet it using mask 255.255.255.128
to create two subnets. This gives us network address ranges and improves network efficiency.

Benefits include better network organization, improved security, and reduced broadcast traffic.